

Controlled Shop Pack

At-machine working set - controlled braking followed by delayed STO and line isolation

Implementation documents

- Wire Schedule Rev J is the point-to-point build authority. Ferrule number equals wire number.
- Ladder Logic Rev J is the CLICK PLUS programming authority.
- This Shop Pack supplies the one-line, safety loop, VFD card, and validation sequence.
- The owner-maintained BOM Rev J is the sourcing authority.
- The SCCR Worksheet Rev J controls release of the power circuit and panel SCCR marking.

Rev J safety and hardware decisions

- Dold BH5928-92-61-24-1: 24 V, fixed screw terminals, 0.1-1.0 s adjustable release delay.
- Instantaneous safety contacts independently open Run-FWD and Run-REV. The GS20 performs a controlled ramp stop.
- Two Phoenix Contact 2966265 modules: CR-S1 asserts DI4; CR-S2 sources PLC X013. Both are advisory and their polarized coils are driven by 31-32.
- Three delayed contacts independently open Contactor A, STO1, and STO2. X011 proves delayed KA status.
- No delayed output remains spare and Rev J includes no mechanical brake.
- Initial drive deceleration is 0.50 s; commission the relay at 1.00 s. Final delay is provisional until measured.

Fuse map

Position	Protects	Holder	Element	Rating
Branch (3x)	VFD input branch	LFT300603C, 3-pole	TJN35 x3	35 A Class T
PSU-FU	208 V PSU tap	DN-F10MN x2	GMC-3 x2	3 A time-delay
KB-FU	Contactor-B tap	DN-F10MN x2	GMC-3 x2	3 A time-delay
24V-FU	+24 V main bus	DN-F10MN	GMC-5	5 A time-delay
SR-FU	Safety-relay feed	DN-F10MN	GMC-3	3 A time-delay

POWER-CIRCUIT PROCUREMENT CHECK PASSED
The LC1D18BD is rated 100 kA with fuses through 40 A. The GS20 manual specifies TJN35 Class T protection for GS23-22P0 and rates GS20 drives for circuits up to 100 kA. Final panel marking still requires available-fault-current confirmation and field sign-off.

STOP-TIME LIMITATION
At 2,000 RPM a linear 0.50 s stop represents about 16.7 spindle revolutions; 0.25 s represents about 8.3. This design materially improves coast-down but is not contact-triggered SawStop behavior.

DWG 01 - Power One-Line

208/240 V 3PH - permanent high-speed motor connection - VFD braking resistor

Main power path

Stage	Device	Build requirement
1. Supply	Rear line-entry j-box	3PH plus PE; remove the front AH7810 switch and line-voltage operator loop.
2. Disconnect	ABB OT30F3	UL 98, 3-pole, 30 A, non-fused. External handle and shaft selected with enclosure. Padlockable OFF.
3. Branch protection	LFT300603C + 3 x TJN35	35 A Class T. Install one LFT30060FBC cover per pole.
4. Contactor A	Schneider LC1D18BD	3 poles, 18 A AC-3 / 32 A AC-1, 24 VDC coil. Held during braking; opens after BH5928 delay. 21-22 mirror N.C. is EDM; 13-14 N.O. is X011.
5. VFD	GS23-22P0	2 HP, 230 V class, 3PH input. DB resistor on + / BR. Separate delayed STO1 and STO2.
6. Motor	Existing 1.5 HP motor	Permanent high-speed lead connection. Shielded U/V/W plus dedicated PE.

Auxiliary power and braking

Function	Device / wiring	Notes
24 VDC control	Owner NDR-240-24	PSU-FU: 2 x GMC-3 on the line tap; 24V-FU GMC-5; SR-FU GMC-3.
Contactor B theater	Vintage Furnas through CR-B 52102	Both line tap legs fused at origin. Main contacts switch no load. Y004 supplies 300 ms chunk.
Dynamic braking	BR-N1-280W50, 50 ohm / 280 W	Operationally essential to rapid stopping. Thermal N.C. to X012. Mount high; internal stirring fan.
PE and shield	Dedicated PE and 360-degree motor-cable clamp	Bond enclosure, panel, door, disconnect, KA, KB, rails, VFD, motor, and machine frame. Target PE continuity below 0.1 ohm.

ORDER OF DEVICES

Line -> OT30F3 -> LFT300603C/TJN35 -> LC1D18BD -> GS23-22P0 -> motor. Disconnect line-side terminals remain live with the switch OFF.

SCCR SIGN-OFF

The selected contactor, fuse, disconnect, and VFD conditions pass the 10 kA target. Confirm available fault current and complete the field sign-off before applying the final panel SCCR label.

DWG 02 - Stop Category 1 Safety Loop

Dual-channel E-stop - monitored reset/EDM - controlled braking - delayed independent safe state

Dold BH5928-92-61-24-1

Circuit	Terminals	Function
E-stop channel 1	S11 -> NC1 -> S12	Positive-opening N.C. contact
E-stop channel 2	S31 -> NC2 -> S32	Positive-opening N.C. contact; cross-fault monitoring configured
Configuration	S21-S22; Y39-Y40	Cross-fault/timing jumpers per exact relay instructions; seal delay after validation
Manual reset / EDM	S34 -> RESET -> KA 21-22 -> S33	KA base-device mirror N.C.; welded KA blocks reset

Output allocation

Timing	Contact	Destination	Role
Instant	13-14 N.O.	Y001 to DI1	Open Run-FWD
Instant	23-24 N.O.	Y002 to DI2	Open Run-REV
Instant monitor	31-32 N.C.	CR-S1 A1+ and CR-S2 A1+	Close on trip; two 9 mA polarized coils
Delayed	47-48 N.O.	LC1D18BD A1+	Open Contactor A after validated delay
Delayed	57-58 N.O.	VFD STO1	Independent torque-removal channel
Delayed	67-68 N.O.	VFD STO2	Independent torque-removal channel

CR-S split relay wiring

Relay	Coil	Contact path	Result
CR-S1 - Phoenix 2966265	A1+ from SR32; A2- to 0 V	VFD DCM -> 11-14 -> DI4	GS20 function 18 Force to Stop
CR-S2 - Phoenix 2966265	A1+ from SR32; A2- to 0 V	+24 V -> 11-14 -> PLC X013	Immediate advisory trip indication

POLARITY

Both Phoenix modules contain input protection and flywheel diodes. Observe A1+/A2- and do not add external suppression. Contacts 12 are unused.

Sequence on E-stop

- Instant contacts open DI1 and DI2. Contact 31-32 energizes CR-S1/CR-S2, asserting DI4 and X013.
- GS20 ramps according to P07.20=2 and P01.15 while KA and both STO channels remain enabled.
- At the validated delay, KA, STO1, and STO2 open independently; KA 13-14 drops X011.
- Reset is blocked until delay completion and healthy EDM. Reset cannot restart the spindle.

VFD Parameter Card - GS23-22P0

Enter with motor uncoupled if practical - parameter numbers per current GS20 manual

Function	Setting	Why
Motor current	6.42 A	Electronic thermal overload
Motor base	60 Hz / measured supply voltage	Use the verified machine supply and motor base point
Command source	External terminals	PLC brokers operational commands
Main / preset 1	60 Hz / 30 Hz	High and low ranges
DI1 / DI2 / DI3	FWD / REV / preset 1	DI1 and DI2 pass through separate instant safety contacts
DI4	Function 18 - Force to Stop	Asserted by CR-S1 11-14 on trip
P00.22	0 - ramp to stop	Never coast on loss of run command
P01.13	0.50 s initial	Normal/fallback deceleration
P01.15	0.50 s initial	Emergency deceleration time 2
P01.26 / P01.27	0.00 s / 0.00 s	Remove deceleration S-curve timing addition
P07.20	2	Force-to-stop uses deceleration time 2
DI function 28	UNUSED	It disables output and free-runs/coasts
DB chopper	Enabled; resistor on + / BR	Absorbs stop and reversal energy
STO1 / STO2	Separate delayed contacts	Independent torque-removal paths
DI mode	NPN/sink - internal power	DCM bonded to CLICK 0 V; never apply +24 V to DCM

Control-wiring checks

- Leave the GS20 input selector at factory NPN/sink. CLICK Y001/Y002/Y003 and dry safety contacts pull inputs to DCM.
- Wire 81 bonds CLICK 0 V to DCM. DCM and STO common share potential; never land external +24 V on DCM.
- Y001 -> 13-14 -> DI1; Y002 -> 23-24 -> DI2. SR31-32 drives both CR-S coils; CR-S1 11-14 closes DCM to DI4; CR-S2 11-14 sources +24 V to X013.
- Set P01.26=P01.27=0.00 s. Factory 0.20 s S-curve values add about 0.20 s to nominal 0.50 s deceleration.
- Initial relay delay is 1.00 s. Reduce only after worst-case measured stop time is <=0.50 s and margin remains.
- A 0.25 s target is aspirational and may be rejected by overvoltage, overcurrent, belt/CVT dynamics, chuck retention, or resistor temperature.

Commissioning - Before Power and VFD Setup

Check in order - LOTO before wiring work - begin motor uncoupled where practical

SCCR SIGN-OFF REQUIRED

LC1D18BD and GS23-22P0 are verified with the selected 35 A TJN35 Class T protection. Confirm available fault current and complete the field sign-off before applying the final panel SCCR label.

Before first power-up

- Continuity and insulation check complete; no line voltage reaches operator controls or machine-front lamps.
- Power order verified: line -> OT30F3 -> LFT300603C/TJN35 -> KA -> VFD. Disconnect line terminals front-protected; three fuse pole covers installed.
- PE continuity below 0.1 ohm including dedicated machine-frame bond, door braid, rails, contactors, VFD, and motor.
- BH5928 inputs wired S11-S12 and S31-S32; S21-S22 and Y39-Y40 jumpers fitted; manual reset/EDM proven by continuity.
- Instant paths: Y001 through 13-14 to DI1 and Y002 through 23-24 to DI2. No bypass around either contact.
- Delayed paths: 47-48 to KA, 57-58 to STO1, 67-68 to STO2. STO factory jumper removed.
- +24 V -> SR31-32 -> CR-S1/CR-S2 A1+; each A2- -> 0 V. CR-S1 11-14 closes DCM to DI4; CR-S2 11-14 sources +24 V to X013.
- LC1D18BD 21-22 N.C. is in EDM; 13-14 N.O. supplies X011; built-in coil suppressor used with correct polarity.
- DB resistor mounted high; thermal N.C. to X012; internal stirring fan operates whenever 24 V is up.

Mechanical checks

- Depth-rod metal stop takes bottom load; neither limit-switch lever is used as a mechanical stop.
- Bottom and top limit tags set slightly before the physical stops.
- Carlo Gavazzi ICF12 target is clean and ferrous; SIO mode selected; X010=1 only with lever confirmed OFF.
- Chuck, arbor, belts, CVT sheaves, guards, and tool retention inspected before any rapid-stop test.

VFD setup - staged

- Measure supply voltage; enter 6.42 A, 60 Hz, and the verified motor base voltage; enable electronic thermal protection.
- Set P00.22=0, P01.13=0.50 s, P01.15=0.50 s, P01.26=P01.27=0.00 s, P07.20=2, and DI4 function 18. Confirm function 28 is unused.
- Enable braking chopper. Verify DB resistor value/wiring and forced-open X012 response before loaded stopping.
- At low speed with light tooling, verify FWD/REV directions and that E-stop commands a ramp rather than coast.
- Increase speed and inertia in stages; stop immediately if the drive trips, belt/CVT shifts, chuck loosens, or resistor overheats.

Commissioning - Safety and Sequences

Do not release for use until every required test passes and results are recorded

Safety sequence and timing

- Set BH5928 delay to 1.00 s. With spindle running, E-stop picks CR-S1/CR-S2 and X013, opens DI1/DI2, and starts a controlled ramp.
- During braking, KA and both STO inputs remain enabled. At delay expiry KA, STO1, and STO2 open independently; X011 falls with KA.
- Measure stop time at minimum and maximum CVT settings, FWD and REV, and maximum expected chuck/tool inertia. Record each result.
- Reduce delay to about 0.75 s only after worst-case stop time is ≤ 0.50 s and at least 0.20-0.25 s margin remains.
- EDM test: prevent KA 21-22 from proving open and confirm reset is refused. Remove the test condition before proceeding.
- After reset, spindle remains stopped and a fresh FWD command is required. E-stop during TAP returns state to IDLE.

PLC indication and sequence

- Immediate trip: X013=1, C20/C60 clear, green turns off, red turns solid even while X011 remains high during braking.
- Final state: X011=0 after KA opens; red remains solid. Valid reset re-energizes KA and restores X011 without restarting.
- DRILL: FWD and REV latch separately; opposite button stops first; second press starts opposite direction; no plug reversal.
- TAP: lever-OFF proof required; bottom reverses; top stops; lever engagement mid-cycle faults and stops.
- JOG: 5 s FWD hold only arms mode; fresh hold-to-run FWD/REV at 30 Hz; drum OFF and any trip inhibit motion.

Acceptance boundary

- SCCR worksheet signed; final panel marking is not greater than the verified available fault current basis.
- No nuisance drive faults, DC-bus overvoltage, or contactor opening before the spindle reaches zero.
- No belt/CVT upset, chuck/arbor release, abnormal motor noise, or DB-resistor thermal trip over the planned stop frequency.
- If ≤ 0.50 s cannot be demonstrated with margin, do not claim the target. Increase delay and revisit braking hardware/drive architecture.
- This is not contact detection and cannot guarantee stopping within one revolution at 2,000 RPM.

Recorded item	Result
Available fault current / final panel SCCR	_____
Worst-case measured stop time	_____
Final BH5928 delay setting	_____
Maximum validated chuck/tool inertia	_____
Reviewed by / date	_____